

Notation, Abbreviations, and Glossary

Notation, Abbreviations, and Glossary I

This appendix consolidates the mathematical notation, abbreviations, hypothesis identifiers, and key terminology used throughout the manuscript. Each table is self-contained and may be consulted independently. Cross-references in the main text use the labels declared here.

Mathematical Symbols and Notation I

The following symbols appear in the methodology, results, and technical appendices. Where a quantity is defined formally, the relevant equation is referenced inline; otherwise the description here is the canonical definition. All probabilities and confidences are real-valued in $[0, 1]$, and all aggregate scores are in $[-1, 1]$.

Table 1: Mathematical symbols and notation used throughout the manuscript. Scoring quantities are defined formally in §3.6 and §??; growth metrics in §??; topic-modeling notation in §??.

Symbol	Description
N	Corpus size after deduplication (total unique papers)
n	Subfield paper count (papers within a single domain category)
$T = y_T - y_0$	Time span in years (used for CAGR)
y_0, y_T	First and last years in the publication window
n_y	Number of publications in year y
$w(a)$	Citation-weighted weight of assertion a : $\log(1 + \text{citations}) \cdot c$
(H)	Aggregate citation-weighted evidence score for hypothesis H , range $[-1, 1]$
(H, t)	Cumulative score for H using only assertions from papers published $\leq t$
$S(H), C(H), A(H)$	Supporting / contradicting / all assertion sets for hypothesis H
c	Assertion confidence reported by the LLM, range $[0, 1]$
d	Assertion direction $\in \{\text{supports, contradicts, neutral}\}$
k	Number of latent topics in NMF factorization
$V \in \mathbb{R}_{\geq 0}^{n \times m}$	TF-IDF document-term matrix (documents $\times$ terms)
$W \in \mathbb{R}_{\geq 0}^{n \times k}$	NMF document-topic factor
$H \in \mathbb{R}_{\geq 0}^{k \times m}$	NMF topic-term factor (overloaded notation; context disambiguates)

Abbreviations and Acronyms Used I

The acronyms below appear at least once in the main text, methods, results, or appendices. Domain-specific shorthands such as the A/B/C taxonomy categories (e.g., A1, A2, B, C1–C5) are documented inline at first use in the field overview and the subfield analyses.

Table 2: Abbreviations and acronyms used in this manuscript, listed alphabetically. Where an acronym names a software package or organization, the canonical reference appears in the bibliography.

Abbreviations and Acronyms Used II

Abbreviation	Definition
AIF	Active Inference
ANGC	Active Neural Generative Coding
API	Application Programming Interface
arXiv	Open-access preprint repository (arxiv.org)
BTAI	Branching-Time Active Inference
CAGR	Compound Annual Growth Rate
CC0	Creative Commons Zero (public-domain dedication)
CI	Continuous Integration
CNM	Conceptual Nexus Model (ResNei)
DCM	Dynamic Causal Modelling
DeSci	Decentralized Science
DOI	Digital Object Identifier
EBM	Energy-Based Model
EFE	Expected Free Energy
FAIR	Findable, Accessible, Interoperable, Reusable
FAIR4RS	FAIR Principles for Research Software
FEP	Free Energy Principle
FEPS	Free Energy Projective Simulation
HITS	Hyperlink-Induced Topic Search (Kleinberg)
IaC	Infrastructure as Code
JSON	JavaScript Object Notation
JSONL	JSON Lines (newline-delimited JSON)
KG	Knowledge Graph
KL	Kullback–Leibler (divergence)
LLM	Large Language Model
MBR	Bayesian Model Reduction
MCMC	Markov Chain Monte Carlo
MIT	Massachusetts Institute of Technology
NLP	Natural Language Processing
NMF	Non-negative Matrix Factorization
ORCID	Open Researcher and Contributor ID
PCA	Principal Component Analysis
PDF	Portable Document Format
POMDP	Partially Observable Markov Decision Process
PROV-O	PROV Ontology (W3C provenance data model)
RBM	Restricted Boltzmann Machine
RDF	Resource Description Framework
ResNei	Research Neighbourhood (cognitive-ergonomic platform)
RL	Reinforcement Learning
SDE	Stochastic Differential Equation
SPARQL	SPARQL Protocol and RDF Query Language

Standard Hypothesis Definitions and Identifiers I

The eight hypotheses below define the evaluation rubric used by the LLM-based assertion extractor (extraction pipeline). Each hypothesis is anchored to its primary domain in the A/B/C taxonomy, but assertions are extracted from any paper whose abstract relates substantively to the claim. Quantitative results across these hypotheses are reported in the hypothesis results section.

Table 3: Standard hypothesis definitions tracked throughout the meta-analysis. The Scope column records the primary domain in the A/B/C taxonomy; assertions are not restricted to that domain. Wording reflects the prompt presented to the extraction LLM.

ID	Hypothesis	Scope
H1	FEP Universality: the Free Energy Principle applies universally to all self-organizing systems, from cells to ecosystems.	A (Core Theory)
H2	AIF Optimality: Active Inference agents achieve principled, near-optimal decision-making under uncertainty by minimizing expected free energy.	B (Tools)
H3	Markov Blanket Realism: Markov blankets correspond to real, physically realizable boundaries between systems and their environments.	A (Core Theory)
H4	Predictive Coding: cortical hierarchies minimize prediction errors via predictive coding, providing a neurobiologically realistic substrate for active inference.	C1 (Neuroscience)
H5	Scalability: Active Inference scales to complex, high-dimensional environments comparable to those addressed by deep reinforcement learning.	B (Tools)
H6	Clinical Utility: Active Inference produces clinically useful computational models of psychiatric and neurological conditions.	C4 (Psychiatry)

Glossary of Key Terms I

The glossary below defines pipeline-specific concepts, statistical methods, and domain terminology referenced in the main text. Software package names appear in typewriter font; mathematical objects use the notation defined above. Where a term has both a colloquial and a technical sense, the technical reading is given.

p4cmp11cm Glossary of key terms used in this manuscript, including pipeline-specific concepts, statistical methods, and domain terminology.

Term	Definition
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Term	Definition
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Active Inference	A framework in which agents minimize expected free energy to select actions, unifying perception, learning, and decision-making under the Free Energy Principle.
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Glossary of Key Terms II

Assertion A directed, confidence-scored claim linking a paper to a hypothesis (supports, contradicts, or neutral). The basic unit of evidence in the knowledge graph; a machine-extracted classification, not a human verdict.

Bayesian Mechanics The formal extension of FEP that grounds Markov-blanket dynamics in stochastic physics, casting belief updates as gradient flows on a free-energy potential.

Canonical ID The unique identifier assigned to each paper during deduplication, following DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > title hash.

Checkpoint A JSON Lines snapshot of LLM extraction progress, recording which papers have been processed and the resulting assertions, enabling incremental resume after interruption.

Citation-Weighted Score The hypothesis-level evidence aggregate combining direction, confidence, and a logarithmic citation weight (Eq. ??).

Glossary of Key Terms III

Compound Annual Growth Rate (CAGR) The constant annual rate that, compounded over the publication window, takes the cumulative count from the first to the last year (Eq. ??).

Conceptual Nexus Model (CNM) The modular knowledge-graph unit used by ResNei; each CNM packages concepts with provenance and supports longitudinal, latitudinal, and relational navigation.

Contrastive Divergence An approximate gradient-based training algorithm for energy-based models [hinton2002training] that truncates the Markov chain used to estimate the gradient of the log-partition function.

Domain Timeline Per-domain yearly publication counts visualizing temporal evolution across the eight tracked categories (A1–A2, B, C1–C5).

Doubling Time (t_d) Years required for cumulative output to double under the prevailing growth rate (Eq. ??).

Glossary of Key Terms IV

Energy-Based Model (EBM) A class of generative models defining $p(x) \propto \exp(-E(x))$ for an unnormalized energy E . Includes Boltzmann machines, Helmholtz machines, and VAEs as special or related cases.

Expected Free Energy (EFE) A scalar combining epistemic (uncertainty-reducing) and pragmatic (goal-achieving) value, minimized over policies. Decomposes equivalently into risk + ambiguity or epistemic + instrumental terms [dacosta2020active].

FAIR Principles Findable, Accessible, Interoperable, Reusable: a set of guiding principles for scientific data infrastructure [wilkinson2016fair]. The pipeline's nanopublications satisfy all four.

Free Energy Principle (FEP) The principle that self-organizing systems minimize variational free energy—an upper bound on surprise—to maintain their structural integrity.

Generative Model A probabilistic model specifying the joint distribution over hidden states and observations, encoding an agent's beliefs about how observations are generated.

Glossary of Key Terms V

Greedy Modularity Maximization The Clauset-Newman-Moore algorithm [clauset2004finding] for community detection.

Implemented via NetworkX `greedy_modularity_communities`; applied here to the citation graph to identify clusters of densely interconnected papers.

HITS Hub/Authority Scores Kleinberg's mutually reinforcing centrality metrics [kleinberg1999authoritative]: hubs point to many authorities; authorities are pointed to by many hubs.

Helmholtz Machine A generative model with separate recognition (bottom-up) and generative (top-down) networks trained by the wake-sleep algorithm [dayan1995helmholtz]; a direct precursor to the variational autoencoder and the FEP's recognition-generation duality.

Incremental Resume The pipeline's ability to continue from where a previous run stopped, loading existing corpus and assertion snapshots and processing only new papers; controlled by `--clear-corpus` and `--clear-assertions` CLI flags.

Glossary of Key Terms VI

Knowledge Graph A directed graph encoding papers, assertions, hypotheses, and their relationships, serialized in an RDF-compatible format.

LLM Config A configuration record specifying the Ollama model name, API URL, sampling temperature, maximum retries, and retry delay used by the assertion extractor.

Markov Blanket A statistical boundary separating internal from external states, defined as the node set that renders a system conditionally independent of its environment.

Mean Year-over-Year Growth ($\bar{g}$) Arithmetic mean of $(n_y - n_{y-1})/n_{y-1}$ across years with non-zero prior-year counts (Eq. ??).

Named Graph An RDF graph identified by a URI, enabling multiple graphs to coexist in a single dataset. Nanopublications use four named graphs (Head, Assertion, Provenance, Publication Info).

Glossary of Key Terms VII

Nanopublication A minimal, self-contained unit of publishable knowledge consisting of an assertion, provenance metadata, and publication context [groth2010anatomy, kuhn2016decentralized].

NMF (Non-negative Matrix Factorization) A factorization $V \approx WH$ with all factors non-negative, used here for unsupervised topic discovery (§??).

Ollama A locally hosted LLM server used for assertion extraction; provides reproducibility and avoids external API dependencies [ollama2024].

PageRank A centrality metric originally designed for web-page ranking. In citation networks, PageRank surfaces influential papers that act as hubs across otherwise disconnected subgraphs.

Precision The inverse variance of a probability distribution; in active inference, precision weighting determines the influence of prediction errors at each level of a hierarchy.

Glossary of Key Terms VIII

Predictive Coding A scheme in which each cortical level passes prediction errors upward and predictions downward, minimizing local free-energy bounds layer by layer.

Progressive Parsing The pipeline's three-stage JSON recovery strategy for malformed LLM output: (1) direct parse, (2) strip Markdown code fences and retry, (3) extract first [...] substring. Papers failing all three are logged and skipped.

Provenance The recorded lineage of an assertion: source paper, extraction model, timestamp, and confidence; serialized in the Provenance named graph of each nanopublication.

Reference Resolution Rate Fraction of all outgoing references that resolve to another paper inside the corpus; reported as 7.4% in the present analysis and used as a lower bound on intra-corpus citation density.

Stochastic Differential Equation (SDE) A differential equation driven by a Wiener (white-noise) process; used in Bayesian-mechanics derivations of Markov-blanket dynamics.

Glossary of Key Terms IX

Surprise (Self-Information) The negative log probability of an observation under the agent's generative model; variational free energy is an upper bound on surprise.

Term Frequency–Inverse Document Frequency (TF-IDF) A weighting that combines normalized term frequency with logarithmic inverse document frequency (Eq. ??); the standard input to NMF in this pipeline.

TriG A serialization format extending Turtle with named-graph support, used to encode nanopublications as RDF datasets.

Trusty URI A URI containing a cryptographic hash of its content [kuhn2014trusty], providing verifiable immutability and content-addressable identification for nanopublications.

Variational Free Energy (VFE) An upper bound on surprise (negative log evidence) decomposable into complexity (KL from prior) and accuracy (expected log-likelihood).

Glossary of Key Terms X

Variational Inference Approximate posterior inference by optimization, replacing intractable marginalization with optimization of a tractable variational distribution.

Ward Linkage A hierarchical clustering method that minimizes total within-cluster variance at each merge step; used to compute domain-centroid dendrograms from mean TF-IDF vectors.

Wong Palette The colorblind-safe 8-color palette of Wong (2011) [wong2011colorblind], used as the standard visualization palette throughout all pipeline-generated figures.